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Analog Circuits — High-Yield Formula Sheet

Small-Signal BJT/MOSFET Models, Op-Amp Circuits, Slew Rate, Feedback & Oscillators

GATE Section: Section 4 • **Marks:** 8 - 11 Marks • **Core Devices:** BJTs, MOSFETs, Op-Amps, 555

OP-AMP & AMPLIFIER GOLDEN RULES
- **Ideal Op-Amp:** Input current I+ = I- = 0. In negative feedback without saturation: V+ = V- (Virtual Short).
- **Slew Rate Limit:** Maximum undistorted output frequency f_max = Slew Rate / (2*pi*V_peak).
- **Feedback Desensitization:** Gain with feedback A_f = A / (1 + A*beta). Bandwidth BW_f = BW * (1 + A*beta).

SECTION: TRANSISTOR SMALL-SIGNAL MODELS & SINGLE-STAGE GAINS

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
BJT Small-Signal Parameters	**g_m = I_C / V_t ~ 40 * I_C (at 300K)** **r_pi = beta / g_m = beta * V_t / I_C** **r_o = V_A / I_C (Early effect)**	I_C = DC collector bias current V_t = 25.86 mV, V_A = Early voltage	Transconductance gm depends ONLY on DC collector current, not transistor geometry!
Common Emitter (CE) Gain	**A_v = -g_m * (R_C || r_o) (without R_E)** **A_v ~ -R_C / (r_e + R_E) ~ -R_C / R_E (with unbypassed R_E)**	Input impedance: R_in = r_pi + (1 + beta)*R_E Output impedance: R_out = R_C	Emitter resistance R_E provides negative feedback: stabilizes gain but reduces gain magnitude.
Common Collector (CC)	**A_v = [(1 + beta)*R_E] / [r_pi + (1 + beta)*R_E] ~ 1**	R_in = r_pi + (1 + beta)*R_E (very high) R_out = R_E || [r_e + R_sig / (1 + beta)] (very low)	Emitter Follower: Unity voltage gain, high input impedance, low output impedance. Used as buffer.
Common Drain (MOSFET)	**A_v = (g_m * R_S) / (1 + g_m * R_S) ~ 1**	R_in = infinity, R_out = R_S || (1 / g_m)	Source Follower: High input impedance, low output impedance.
Miller's Theorem	**C_in = C_f * (1 - A_v)** **C_out = C_f * (1 - 1/A_v)**	C_f = feedback capacitance between in and out A_v = inverting gain (-|Av|)	In inverting amplifiers (A_v < 0), input capacitance is multiplied by (1 + |Av|), severely limiting high-frequency bandwidth.

SECTION: OPERATIONAL AMPLIFIERS & SPECIALIZED CIRCUITS

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Inverting & Non-Inverting	**Inverting: V_out = -(R_f / R_1) * V_in** **Non-Inverting: V_out = (1 + R_f / R_1) * V_in**	Inverting R_in = R_1 (virtual ground) Non-inverting R_in = infinity	Gain of non-inverting amplifier is always >= 1.
Difference Amplifier & CMRR	**V_out = (R_2 / R_1) * (V_2 - V_1) (if R2/R1 = R4/R3)** **CMRR = |A_d / A_cm|, CMRR_dB = 20*log10(CMRR)**	Balanced bridge resistor matching ensures A_cm = 0	Instrumentation amplifier uses two input buffers to ensure CMRR is independent of source resistance.
Op-Amp Integrator	**v_out(t) = -(1 / (R*C)) * integral_0^t v_in(tau) dtau + v_out(0)** **H(s) = -1 / (s*R*C)**	Acts as active low-pass filter	Practical integrator requires large shunt resistor Rf across C to limit DC gain and prevent saturation from input offset.

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Schmitt Trigger Trip Points	**V_UTP = [R1 / (R1 + R2)] * (+V_sat) + [R2 / (R1 + R2)] * V_ref** **V_LTP = [R1 / (R1 + R2)] * (-V_sat) + [R2 / (R1 + R2)] * V_ref**	Hysteresis: V_H = V_UTP - V_LTP	Positive feedback comparator! Eliminates false switching caused by input noise.
Precision Half-Wave Rectifier	**V_out = -V_in for V_in < 0 (diode on)** **V_out = 0 for V_in > 0 (diode off)**	Diode in feedback loop eliminates 0.7V cut-in voltage drop	Can rectify millivolt signals accurately without threshold distortion.

SECTION: OSCILLATORS, TIMERS & FEEDBACK TOPOLOGIES

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Barkhausen Criterion	**|A(j*omega) * beta(j*omega)| = 1** **angle(A * beta) = 0 or 360 deg**	Loop gain magnitude must equal unity Total phase shift around loop must be 0 or 2*pi	Necessary condition for steady sinusoidal oscillations without external input.
RC Phase-Shift Oscillator	**f0 = 1 / (2 * pi * R * C * sqrt(6))** **Required Op-Amp Gain: |A_v| >= 29**	Three identical RC high-pass sections, each contributing 60 deg phase shift	For BJT RC phase-shift with current feedback: h_fe >= 44.5 + 4*(Rc/R) + 23*(R/Rc).
Wien-Bridge Oscillator	**f0 = 1 / (2 * pi * R * C)** **Required Gain: A_v >= 3 (R_f / R_1 >= 2)**	Lead-lag bridge network contributes zero phase at resonance	Frequency can be smoothly tuned over a wide range using ganged variable capacitors.
555 Astable Multivibrator	**f = 1.44 / ((R_A + 2*R_B) * C)** **Duty Cycle = (R_A + R_B) / (R_A + 2*R_B) * 100%**	T_high = 0.693 * (R_A + R_B) * C T_low = 0.693 * R_B * C	Duty cycle is always > 50%. A diode across R_B allows duty cycles < 50%.
555 Monostable Multivibrator	**T_pulse = 1.1 * R_A * C**	Triggered by negative pulse < Vcc / 3	Generates single stable output pulse of precise duration T.